

INTRODUCTION TO COMPUTING

Overview of Computing Systems

Lecture Part 1-1

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<https://apvinod.github.io>



Module Objectives and Learning Outcomes

- **Objectives**

- To provide the foundation skills for subsequent modules taught in the ICT programme, as well as to instill an ICT mindset

- **Learning outcomes**

- Interpret binary numbers used for representing how information are stored in computers.
- Design combinatorial circuits and express the circuits using Boolean algebra.
- Create algorithms for solving simple problems.
- Describe different software engineering methodologies.
- Apply high level design of a system using UML.
- Design and model databases using ER modelling concepts.
- Modify database using simple SQL statements.
- Explain basic cryptography and how it is used to secure computer networks.

Teaching Team



Vinod Prasad
(First half)

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Kent Lam
(Second half)

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Assessments

- **Laboratory Assessments [25%]:**
 - Students to work on their own or in team, the assessments based on the lab will be assessed individually.
 - Lab 1: Number systems and digital circuits [8%]
 - Lab 2: Software Engineering [8%]
 - Lab 3: Database ER/SQL [9%]
- **ICT Special Topic Presentation [15%]:**
 - Prerecorded Presentation (PPT) in Auto-play mode to be submitted on LMS.
Pre-assigned team of 3-4 students, on an Advanced Topic in ICT (Students can form teams and the team to select a topic from the given list of several topics)
- **Quiz [10%]:**
 - Progressive quizzes at the end of lectures [5% each for Part 1 and 2]
- **Summative Test [50%]:**
 - Test 1 (Part 1): Summative test on week 6 [25%]
 - Test 2 (Part 2): Summative test on week 13 [25%]

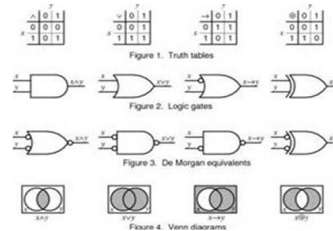
Module Outline



Basics of
Computing,



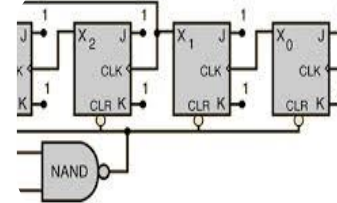
Software
Engineering



Boolean
Logic



Algorithms &
Databases

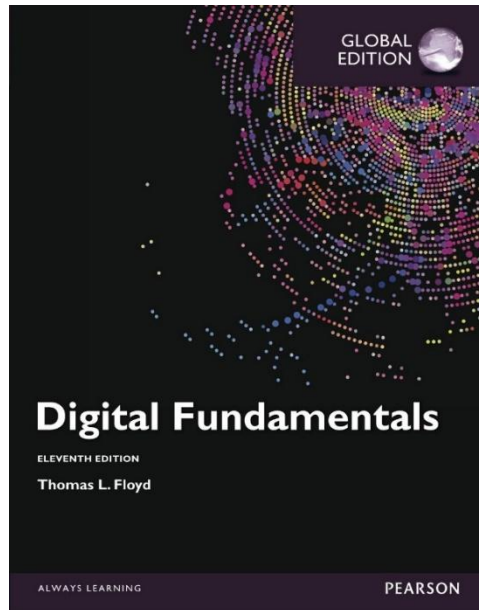


Digital
Circuits

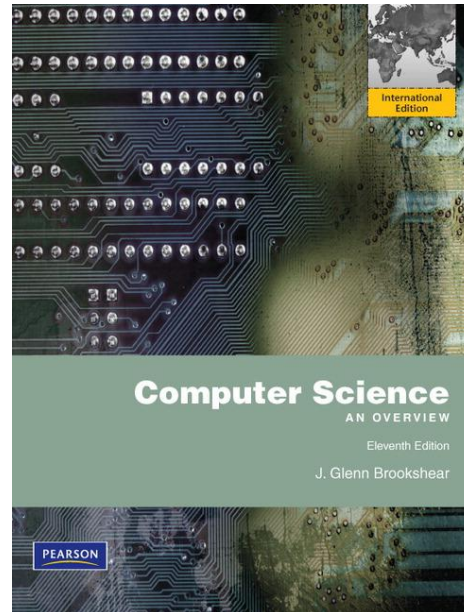


Security

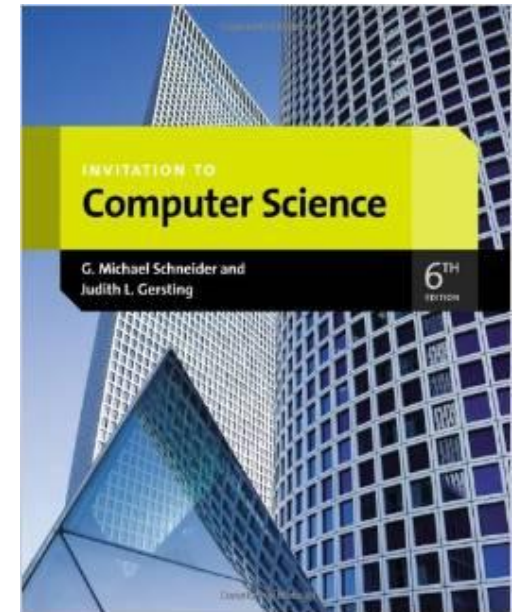
References



Thomas L. Floyd, “Digital Fundamentals”, 11th edition, Pearson International Edition 2015. (e-book available on SIT library online)



Glenn Brookshear and Dennis Brylow, “Computer Science: an Overview”, 12th edition, Pearson Education 2015.



G. Michael Schneider and Judith Gersting, “Invitation to Computer Science”, 6th edition, Boston, MA: Cengage Learning 2013.

Introduction

- Introduction to digital world & ICT
- Digital versus analogue world
- Digitization
- Integrated circuit (IC) revolution
- Moore's Law

Digital Systems in Daily Life



Entertainment systems

Embedded systems

WWW

Biomedical



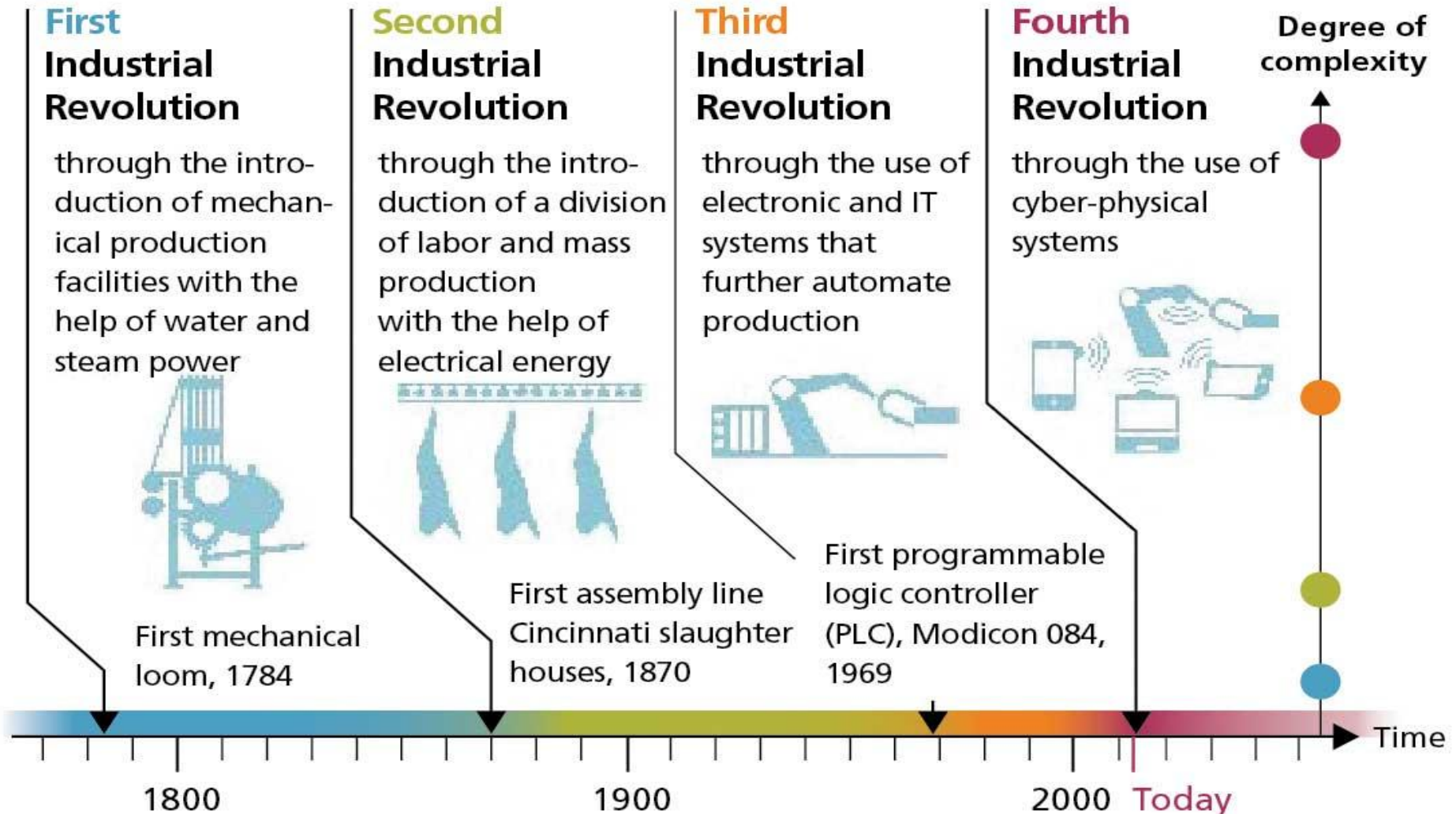
Automotive electronics

Aircraft electronics

Artificial Intelligence

Robotics & Automation

Industrial 4.0



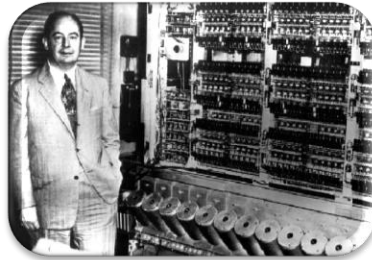
Evolution of Computing Machines



1946 ENIAC (Electronic Numerical Integrator and Computer), Univ. Pennsylvania

Vacuum tube, 63m², 30 tons

Program by physical wiring



1950 EDVAC (Electronic Discrete Variable Automatic Computer)

Vacuum tube
45m², 7850kg



1959-1965 2nd Generation Mainframe Computer (e.g. IBM 7094)
Transistor



1965-1971
3rd Generation Mini Computer
Integrated Circuit (IC)



1971-now
4th Generation Micro Computer
Very Large Scale Integration (VLSI)

What is a Computer

A **computer** is a programmable machine designed to automatically carry out a sequence of arithmetic and/or logical operations.

Types of Computer

- Supercomputer
- Personal computer
- Portable computer (laptop, tablet, smartphone)
- Embedded computer



Supercomputer

Multiple CPUs,
Parallel
processing, High-
speed, Massive
data processing



Embedded computer



Personal computer



Portable computer

What is Information Communication Technology (ICT)?

- A diverse set of technological tools and resources used to create, transmit, receive, process, store, or share information.
- Include computers, Internet, live broadcasting technologies (radio, television and webcasting), recorded broadcasting technologies (podcasting, audio and video players, and storage devices) and telephony (fixed or mobile, satellite, video-conferencing, etc.).

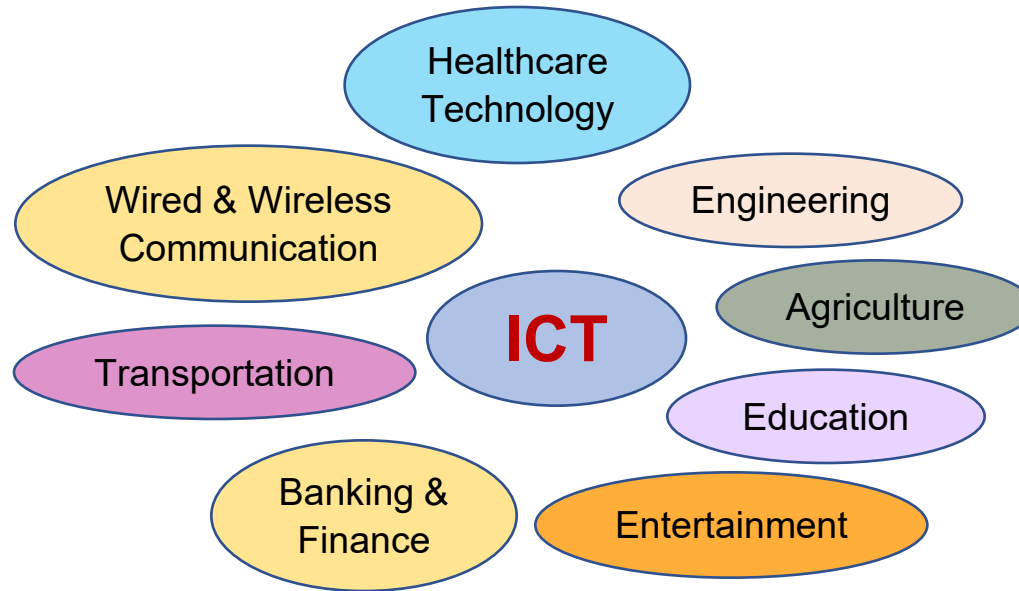
ICT can generally be **classified** into:

Hardware:

- Design and development of information processing hardware.
- e.g. CPU, GPU, digital ASIC, embedded computers, network, server, supercomputer, etc.

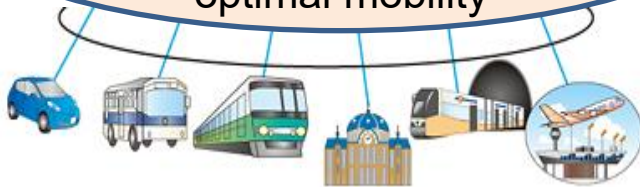
Software:

- Design and development of computer programs to run the hardware.
- e.g. firmware (for embedded systems), operating systems (Windows, Linux).
- Applications: high-level (C++, Java, scripts), databases, web-based, internet, mobile.



ICT in Transportation

Seamlessly connect and coordinate various means of transport and support optimal mobility



Internet of Things (IoT) in Agriculture

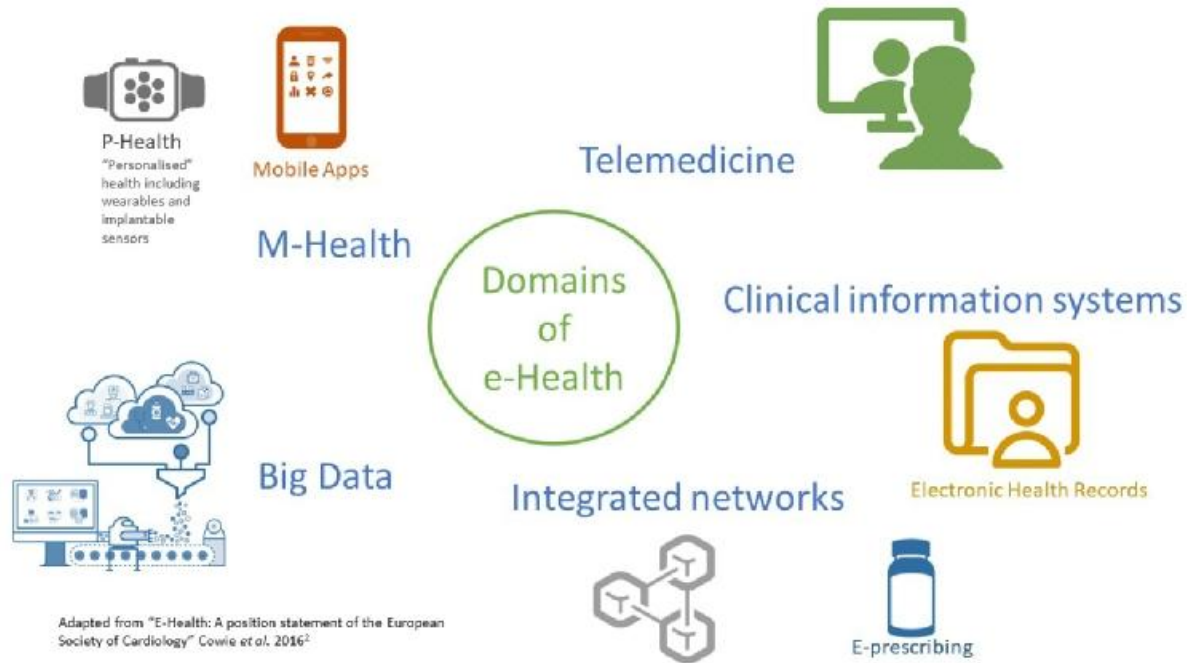
Today, IoT has many use cases in Agriculture



ICT Applications

SIT Internal

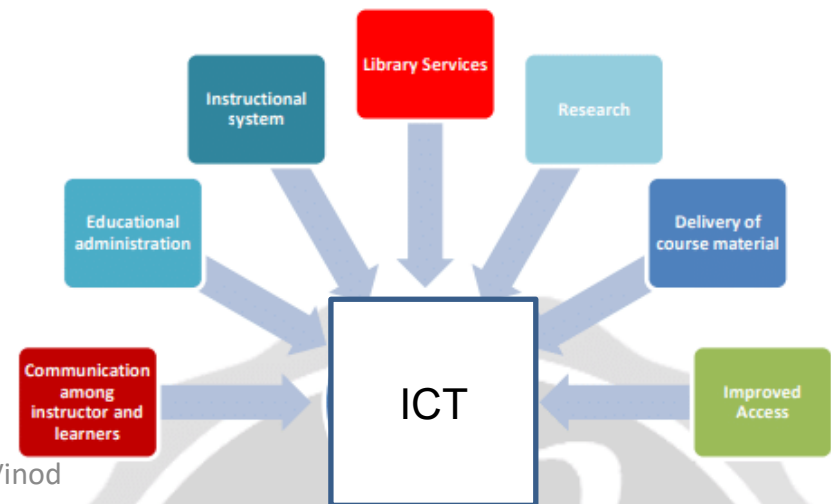
ICT in Healthcare



ICT in Banking



ICT in Education



ICT Construction Engineering



IOT IN CONSTRUCTION INDUSTRY

SITE MONITORING

Construction managers can have real-time insights on employees and machines

MACHINE CONTROL

IoT sensors can guide the machines with greater precision and minimal human involvement

CONSTRUCTION SAFETY

Combining other technologies like AI with IoT, safety measures can be further enhanced, thereby assuring the safety and health of workers

FLEET MANAGEMENT

IoT fleet management solutions can be used for optimizing transit routes and maintaining vehicles

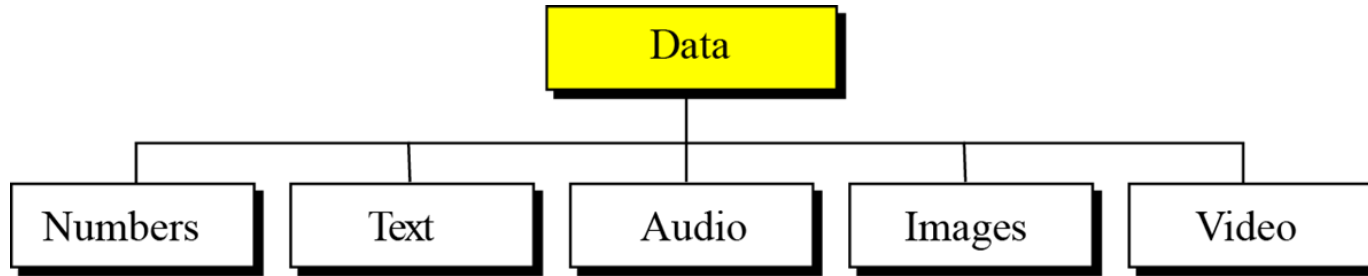
PROJECT MANAGEMENT

IoT devices can facilitate cost-cutting by using site monitoring techniques, keeping the project budget-friendly

ICT is all pervasive in industry and daily life!

How Information (or Data) is Represented in Digital Form?

Information or Data is the representation of an idea/message in the form of an ordered sequence of symbols.



Data stored in a computer is represented as strings of 1's and 0's – **binary numbers**.

Binary Number - Base 2 number representation

- A base 2 digit (0 or 1) is called a **bit**.
- Bit '0' represents 0 volts and Bit '1' represents 5 volts.
- Bit '0' (i.e., 0 volts) is **logic LOW** and Bit '1' (i.e., 5 volts) is **logic HIGH**.

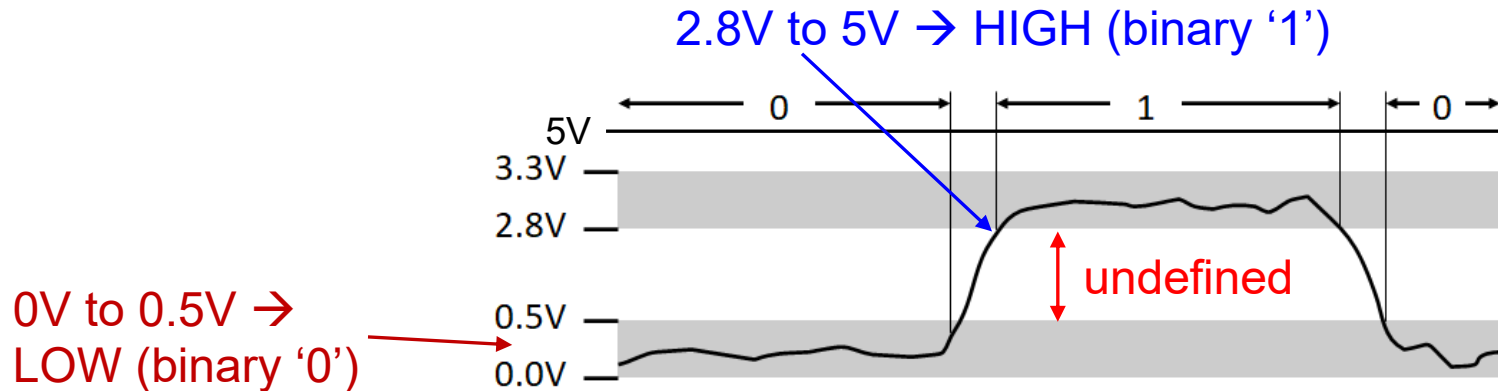
Decimal Number - Base 10 number representation (digits are 0 to 9).

Binary Number Representation

- What are the advantages?

Easy to store with bi-stable elements.

Reliably transmitted on noisy environment.



Why there is a range of voltages for 'LOW' and 'HIGH'?

- To enable Noise Tolerance.

What if the voltage value lies between 0.5V and 2.8V?

- Treated as undefined state.

So it is necessary to ensure that the maximum value of logic LOW is not more than 0.5V and the minimum value of logic HIGH is not less than 2.8V

Binary Number Representation

Binary is a **weighted number system**. The weight of a bit depends on its position (positional weights).

Decimal is also a **weighted number system**. The weight of 0-th digit is 10^0 , 1-st digit is 10^1 , 2-nd digit is 10^2 and so on.

Example: 264_{10} (read as decimal-264) = $4 \times 10^0 + 6 \times 10^1 + 2 \times 10^2 = 264_{10}$.

For Binary, the weight of 0-th bit is 2^0 , 1-st bit is 2^1 , 2-nd bit is 2^2 and so on.

Example: 264_{10} (read as decimal-264) = $4 \times 10^0 + 6 \times 10^1 + 2 \times 10^2 = 264_{10}$.

Positional weights: $2^x \rightarrow$ value depends on the position 'x'.

2^{15}	2^{14}	2^{13}	2^{12}	2^{11}	2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
0	1	0	1	1	0	0	0	1	1	1	0	1	1	0	1

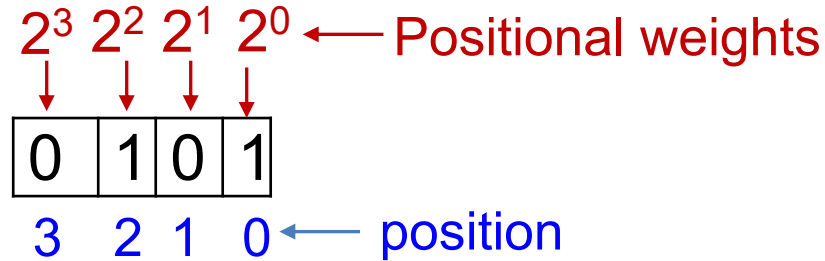
Most Significant Bit (MSB)

Least Significant Bit (LSB)

MSB has the maximum weight whereas **LSB** has the minimum (least) weight.

Binary Number Representation - Examples

Example: Decimal number 5_{10} in Binary is: 0101_2



$$0101_2 = 2^0 \times 1 + 2^1 \times 0 + 2^2 \times 1 + 2^3 \times 0 = 1 + 0 + 4 + 0 = 5_{10}$$

Example: 1010_2

$$1010_2 = 2^0 \times 0 + 2^1 \times 1 + 2^2 \times 0 + 2^3 \times 1 = 0 + 2 + 0 + 8 = 10_{10}$$

Try this Example: 10011010_2

$$10011010_2 = 2^0 \times 0 + 2^1 \times 1 + 2^2 \times 0 + 2^3 \times 1 + 2^4 \times 1 + 2^5 \times 0 + 2^6 \times 0 + 2^7 \times 1 = 0 + 2 + 0 + 8 + 16 + 0 + 0 + 128 = 154_{10}$$

Binary Number Representation – Largest ' n ' bit number

Biggest possible 8-bit value = $\underline{11111111_2} = \underline{255}_{10}$

Smallest possible 8-bit value = $\underline{00000000_2} = \underline{0}_{10}$

Biggest possible 16-bit value = $\underline{1111111111111111_2} = \underline{65,535}_{10}$

Biggest possible n -bit value = $2^n - 1$

Question: What is the least number of binary bits required to represent the decimal number 20?

$$20_{10} = 10100_2 \rightarrow 5 \text{ bits}$$

Question: What is the least number of binary bits required to represent the decimal number 32?

$$32_{10} = 100000_2 \rightarrow 6 \text{ bits}$$

All the above represent 'unsigned' numbers.

How do you represent negative numbers?

How are Negative Numbers Represented in Binary?

Simplest way – Use a ‘sign’ bit:

MSB is the sign bit (negative if MSB=1, positive if MSB=0)

Example: $+12_{10} = \underline{0}1100_2$ whereas $-12_{10} = \underline{1}1100_2$

Another way: Two’s complement (more often used)

Example: $+12_{10} = 01100$ – same as above

To represent -12 in 2’s complement, flip (invert) all bits and add ‘1’

Flip all bits of 01100 $\rightarrow$ 10011. Then add ‘1’ $\rightarrow$ 10100

So, for negative numbers in 2’s complement, just flip all the bits and add 1

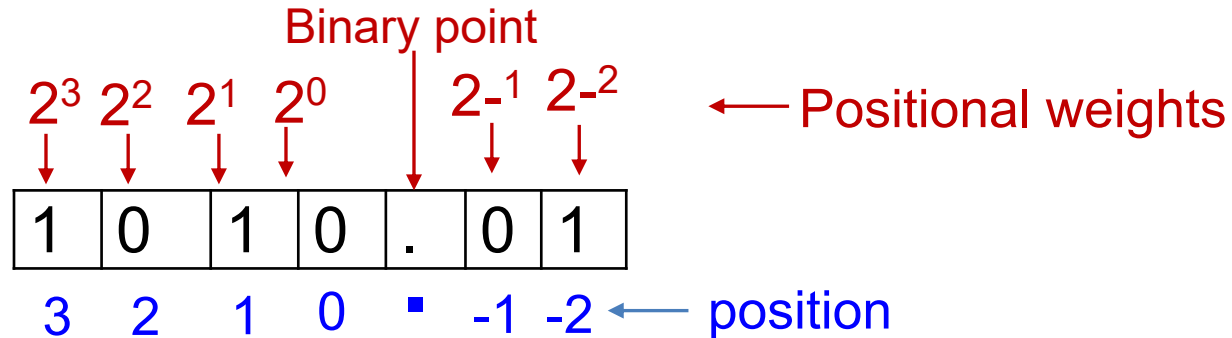
There is nothing intrinsically correct about one system over another.

Either 11100 or 10100 can be used to represent -12, it just depends on what system of interpretation is used.

That is, a human programmer chooses the meaning of the bits.

How are Fractional Numbers Represented in Binary?

Example: 1010.01_2



$$1010.01_2 =$$

$$2^{-2} \times 1 + 2^{-1} \times 0 + 2^0 \times 0 + 2^1 \times 1 + 2^2 \times 0 + 2^3 \times 1 + =$$

$$0.25 + 0 + 0 + 2 + 0 + 8 = 10.25_{10}$$

General Binary representation: $a_n a_{n-1} a_{n-2} \dots a_1 a_0 . a_{-1} a_{-2} \dots a_{-m}$

• Value:

$$a_n \times 2^n + a_{n-1} \times 2^{n-1} + \dots + a_3 \times 2^3 + a_2 \times 2^2 + a_1 \times 2^1 + a_0 \times 2^0$$

$$+ a_{-1} \times 2^{-1} + a_{-2} \times 2^{-2} + \dots + a_{-m} \times 2^{-m}$$

$$= \sum_{i=-m}^n a_i \times 2^i$$

Data (Signal) Representation – Analog versus Digital

- Real-world signals are analog (continuous in time and amplitude).
- In digital systems, data is stored exclusively in a digital format (1s and 0s).
- Need for Analogue to Digital Converter (ADC) and Digital-to-Analog Converter (DAC).

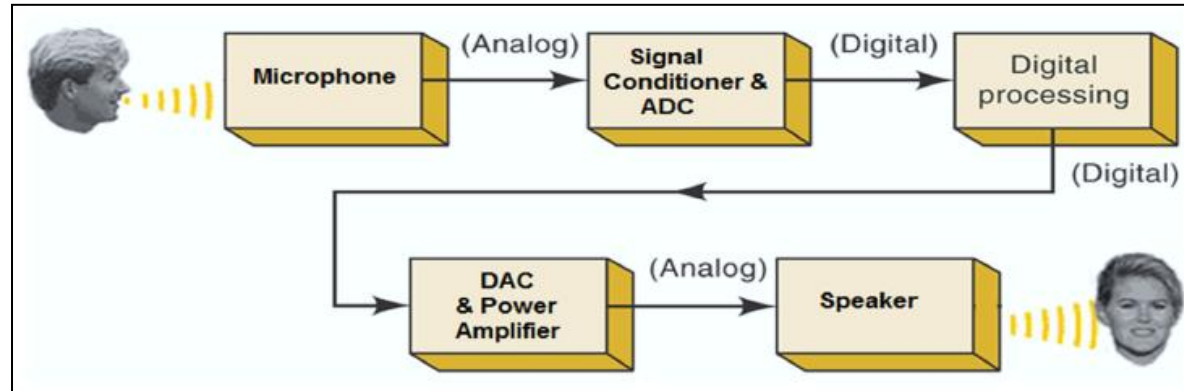
Analog



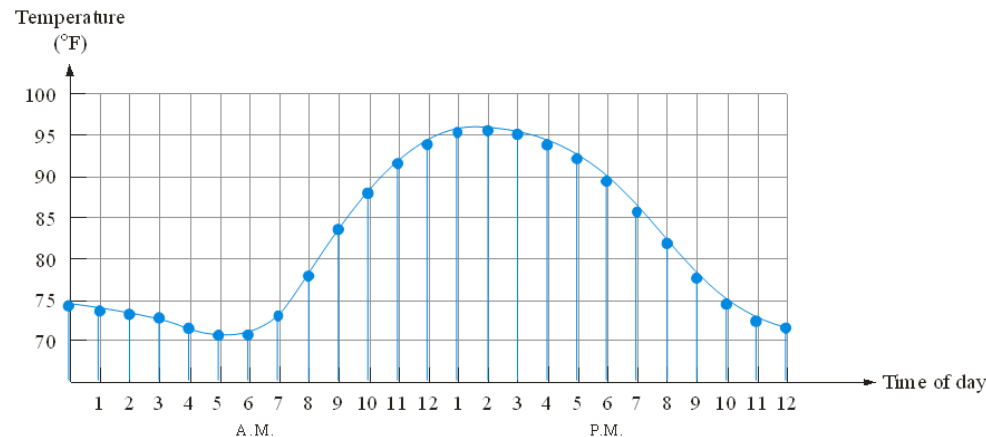
Digital



Analog-to-Digital and Digital-to-Analog Conversions



Physical signals (e.g. temperature) are **analog** and vary continuously in both axes (time and amplitude).



Digital systems can process, store, and transmit data more efficiently but only sample at discrete times (x-axis) and store discrete values (y-axis).

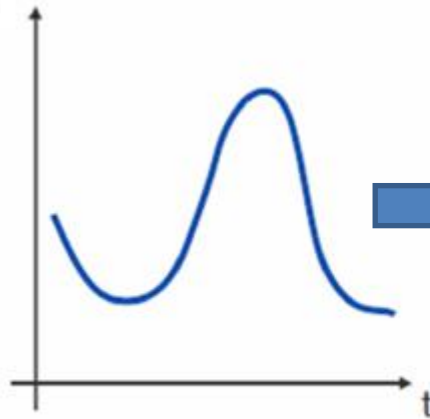
An **analog-to-digital conversion** is a 2-step processes:

- 1. **Sampling**: Using a sample and Hold circuit
- 2. **quantization**: Using an analog to digital converter (ADC)

Data (Signal) Representation – Analog versus Digital

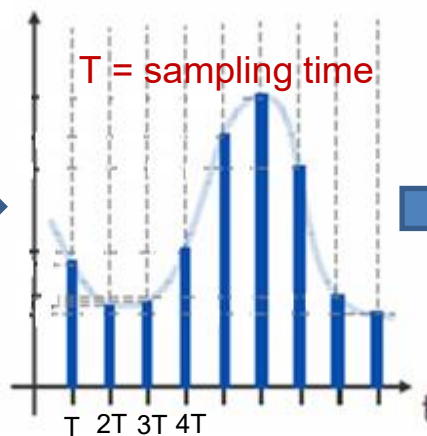
Step 1: Sampling (x-axis) **Step 2**: quantization (y-axis)

amplitude



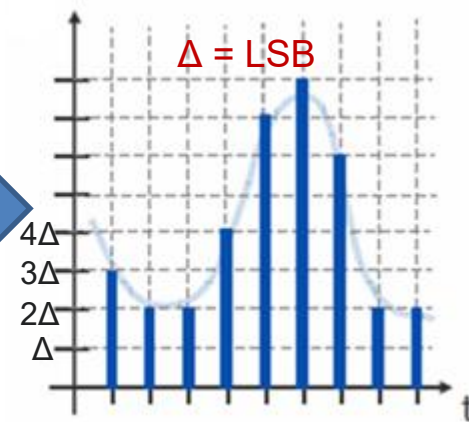
Analog signal: continuous-time, continuous-amplitude

amplitude



Sampled signal: discrete-time, continuous-amplitude

amplitude



Digital signal: discrete-time, discrete-amplitude

Analog-to-Digital Conversion: Sampling & quantization

Analog Signal:

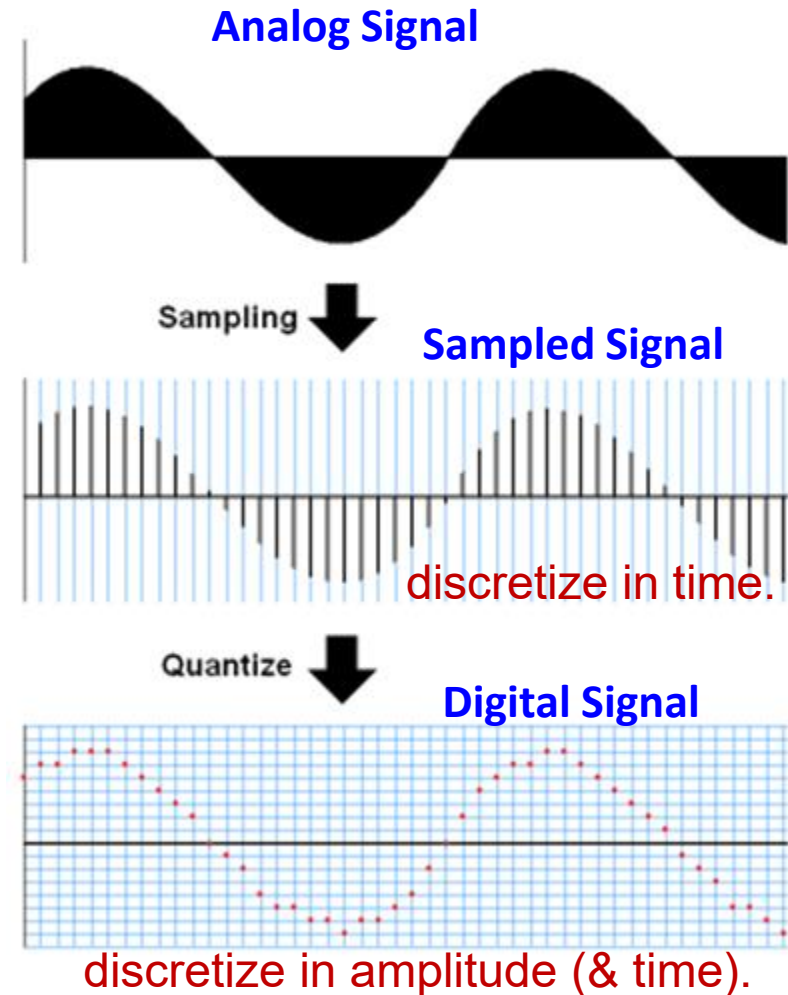
- Continuous in time and amplitude.

Step 1: Sampling:

- A process to convert a continuous-time signal into a discrete-time signal.
- This deals with the regular interval measurement on the **x-axis** (time axis)

Step 2: quantization:

- The process of mapping continuous amplitude values to discrete output values.
- This deals with the **y-axis** where the amplitude is divided into equal steps and each sample have to take on one of the defined step value.



Signal Sampling

How often (fast) should we sample a signal?

- How to choose sampling frequency (rate).

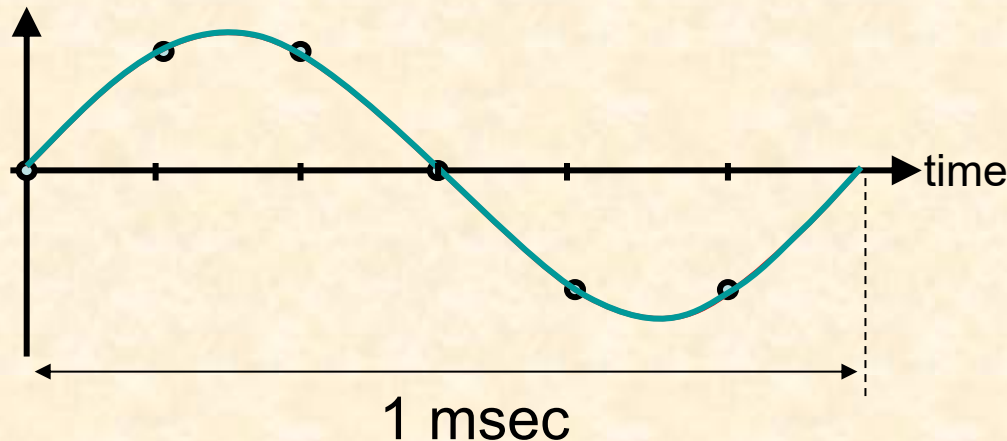
How many data (samples) do we need to take from the signal?

Sampling of Signal

Example 1:

Frequency of signal $f_{SIG} = 1 \text{ KHz}$ (1 cycle in 1 msec)

Sampling frequency $f_s = 6 \text{ KHz}$ (6 samples in 1 msec)



Reconstructed signal frequency (Observed frequency)
= 1 KHz (OK)

Ratio: $f_s/f_{SIG} = 6/1 = 6$

Sampling of Signal (cont)

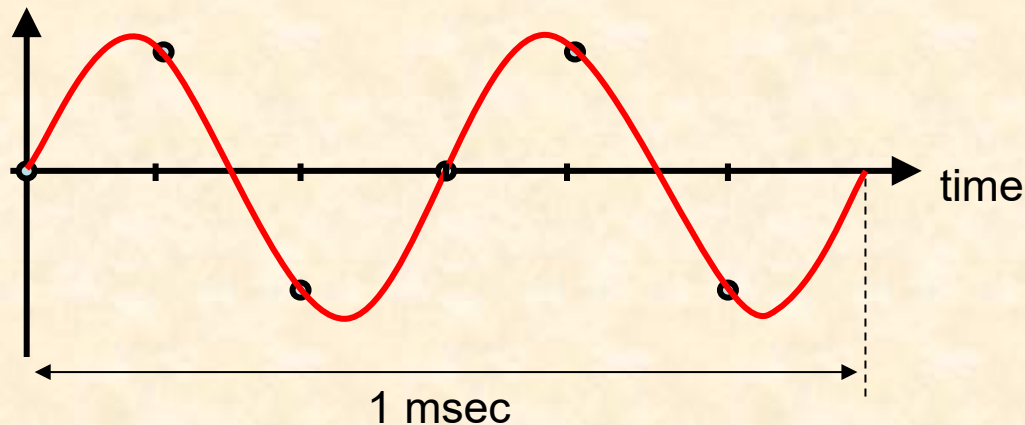
Example 2: (further reduced in samples per cycle)

Frequency of signal $f_{SIG} = 2 \text{ KHz}$

- 2 cycles in 1 msec

Sampling frequency $f_s = 6 \text{ KHz}$

- 6 samples in 1 msec



Observed frequency = 2 KHz (OK)

Ratio: $f_s/f_{SIG} = 6/2 = 3$

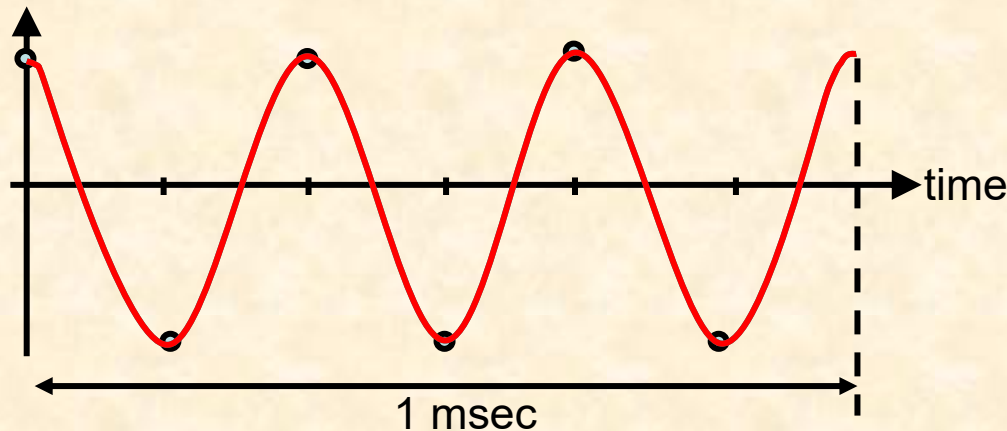
Sampling of Signal (cont)

Example 3:

Frequency of signal $f_{\text{SIG}} = 3 \text{ KHz}$

- (3 cycles in 1 msec)

Sampling frequency $f_s = 6 \text{ KHz}$



Observed frequency = 3KHz (OK)

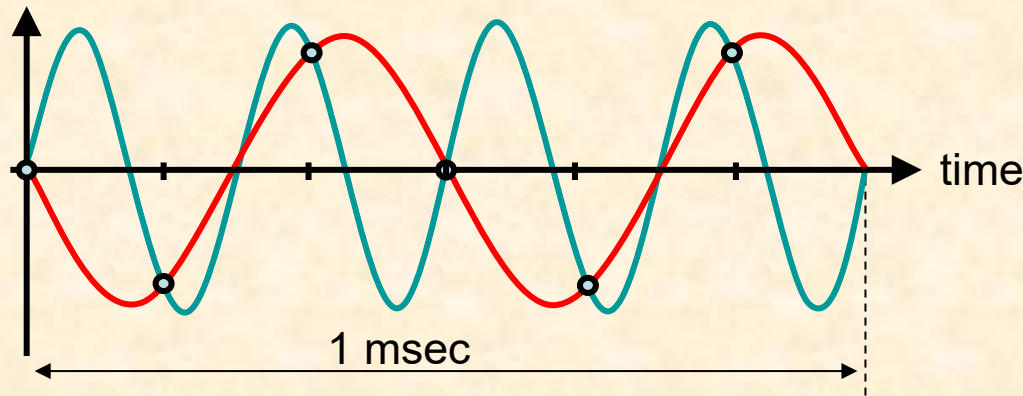
Ratio: $f_s/f_{\text{SIG}} = 6/3 = 2$

Sampling of Signal (cont)

Example 4:

Frequency of signal $f_{SIG} = 4 \text{ KHz}$ (4 cycles in 1 msec)

Sampling frequency $f_s = 6 \text{ KHz}$ (6 samples in 1 msec)



Observed frequency = 2 KHz! (i.e. **wrong; reconstructed signal is different from the original signal $\rightarrow$ aliasing occurred**)

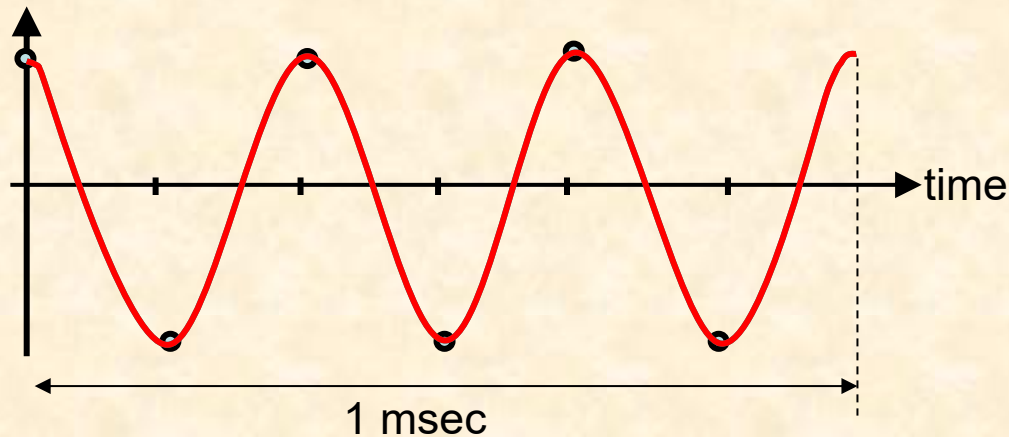
Ratio: $f_s/f_{SIG} = 6/4 = 1.5$

What should be the Minimum Sampling Frequency?

Example 5: (Example 3 where $f_s/f_{SIG} = 2$)

Frequency of signal $f_{SIG} = 3$ KHz (3 cycles in 1 msec)

Sampling frequency $f_s = 6$ KHz (6 samples in 1 msec)



Observed frequency = 3KHz i.e. correct freq

(Ratio: $f_s/f_{SIG} = 6/3 = 2$)

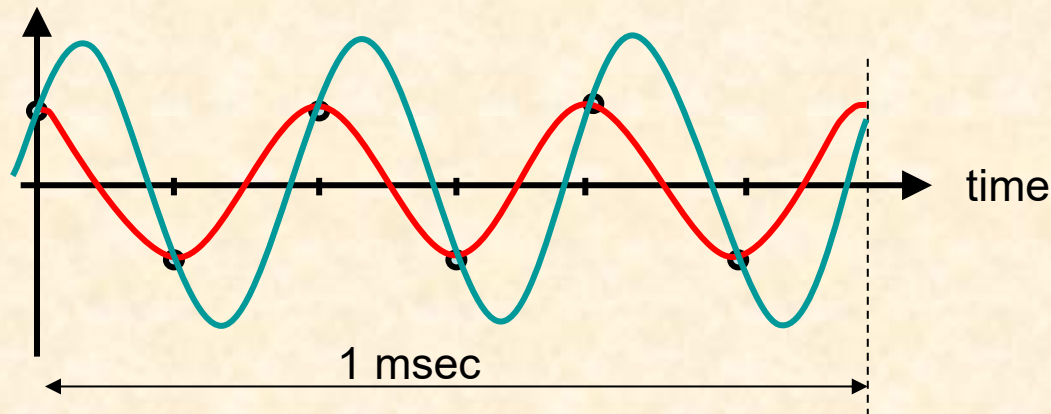
Minimum Sampling Frequency (cont)

Example 6:

Frequency of signal $f_{SIG} = 3 \text{ KHz}$ (3 cycles in 1 msec)

Sampling frequency $f_s = 6 \text{ KHz}$ (6 samples in 1 msec)

Ratio: $f_s/f_{SIG} = 6/3 = 2$



Observed frequency = 3KHz

i.e. correct frequency but wrong amplitude & phase (because sampling is not done in phase with signal).

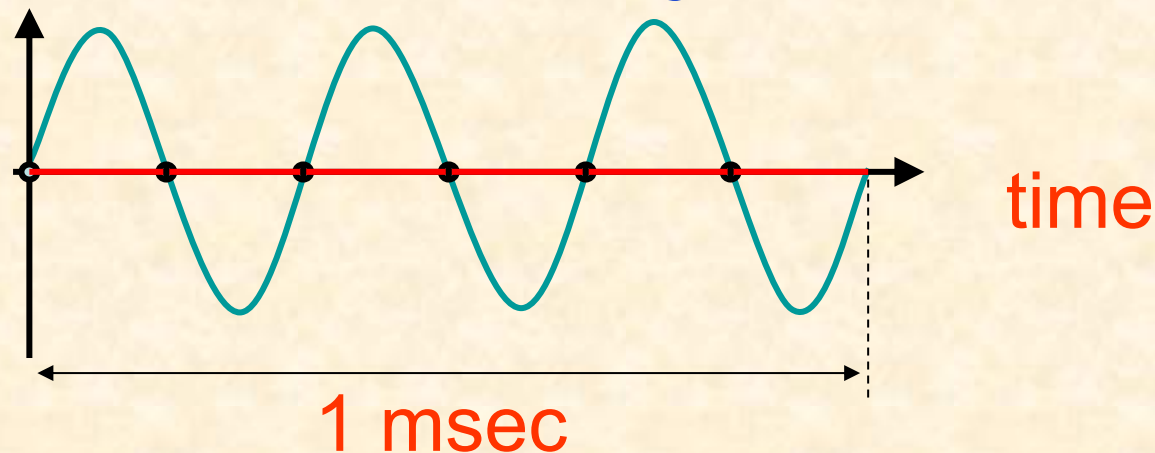
In this case, it is safer to fix f_s/f_{SIG} slightly larger than 2.

Minimum Sampling Frequency - *critically sampled*

Example 7:

Frequency of signal $f_{\text{SIG}} = 3 \text{ KHz}$

Sampling frequency $f_s = 6 \text{ KHz}$



Observed frequency = 0Hz!

$$\text{Ratio: } f_s / f_{\text{SIG}} = 6/3 = 2$$

Nyquist-Shannon Sampling Theorem

Harry Nyquist (1889-1976, Engineer @ Bell Labs) and
Claude Shannon (1916-2001, Engineer @ Bell Labs)

Shannon's Mouse ("Theseus") - First Artificial Intelligence Device (1950) !

Maze navigation: Magnetic mouse whose movement controlled by electromechanical relay circuit.

The mouse was designed to search through the corridors until it found the target.

The mouse could be placed anywhere it had been before, and because of its prior experience it could go directly to the target.

If placed in unfamiliar territory, it was programmed to search until it reached a known location and then it would proceed to the target, adding the new knowledge to its memory and learning new behaviour.



Sampling Theorem

A bandlimited continuous-time signal, with highest frequency (bandwidth) F_{max} , can be uniquely reconstructed from its samples provided that the sampling rate $F_s \geq 2 F_{max}$.

Sampling frequency must be

- **At least twice the signal bandwidth:**

$$F_s \geq 2F_{max}$$

i.e. Signal bandwidth must be

- **less than half the sampling frequency:**

$$F_{max} \leq \frac{1}{2} F_s$$

Nyquist Frequency

Nyquist Frequency

- highest frequency of a signal that can be sampled properly (without causing aliasing), i.e., F_{max} .
- equal to half the sampling frequency, i.e., $F_s / 2$.

Nyquist Rate

- The minimum sampling rate required to avoid aliasing of signal, i.e., F_s .
- i.e. Nyquist rate is twice the highest frequency of the signal, $F_s = 2 F_{max}$.
- A signal is
 - *under-sampled* if sampled below the Nyquist rate
 - *critically sampled* if sampled at Nyquist rate
 - *over-sampled* if sampled higher than the Nyquist rate

Aliasing occurs if sampling theorem is not satisfied

- An aliased signal provides a **poor representation** of the analog signal.
- Aliasing is an effect that causes different signals to become indistinguishable (or *aliases* of one another) when sampled.
- Aliasing causes **false frequency/amplitude/phase** component to appear in the reconstructed signal.
- It is not possible to reconstruct the original signal from the signal samples once aliasing occurs.

Sampling theorem

Critical thinking...

When sampled with Nyquist rate (at the border line), can you find a case of

- Wrongly observed frequency?
- Correctly observed frequency but wrong amplitude & phase?



What should the conclusion be?

- Nyquist sampling rate: the necessary (but not the sufficient) condition for signal reconstruction
 - It is desirable to use a sampling rate $F_s > 2 F_{max}$ (instead of $F_s = 2 F_{max}$).

Practical Sampling rates

Signal Type	Signal Bandwidth	Minimum Sampling Rate	Rate Actually Used
Telephone-quality speech	3.5 kHz	7000 samples/s	8000 samples/s
Music	20 kHz	40,000 samples/s	44,100 samples/s
FM radio	200 kHz	400,000 samples/s	500,000 samples/s
Standard-definition television	6 MHz	12,000,000 samples/s	14,400,000 samples/s

- Signal bandwidth refers to the difference between the highest frequency in the signal and the lowest frequency in the signal.
- If the lowest frequency is zero, then the signal's bandwidth is equal to the highest frequency.

Daily-life examples of aliasing

Human eye has a persistence of $1/10^{\text{th}}$ of a second, i.e. eye can capture and differentiate only up to 10 items/events per second. This is a limitation of our eye's sampling frequency!

- A fan rotating in clockwise direction at very high speed 'appears' as if it is rotating anti-clockwise at a lower speed than the original speed.

Is Aliasing always bad?

- **Aliasing is exploited in movie (video) technology:**
A movie is filmed at 24 frames per second, which is more than twice the bandwidth (frequency) of human eye. Thus eye is unable to differentiate each frame; appears as 'continuous motion'. **So aliasing is not always bad!**

Signal Quantization

When an analog signal is converted into digital form

- loss of signal resolution
- **Quantization** of signal into multilevel signals

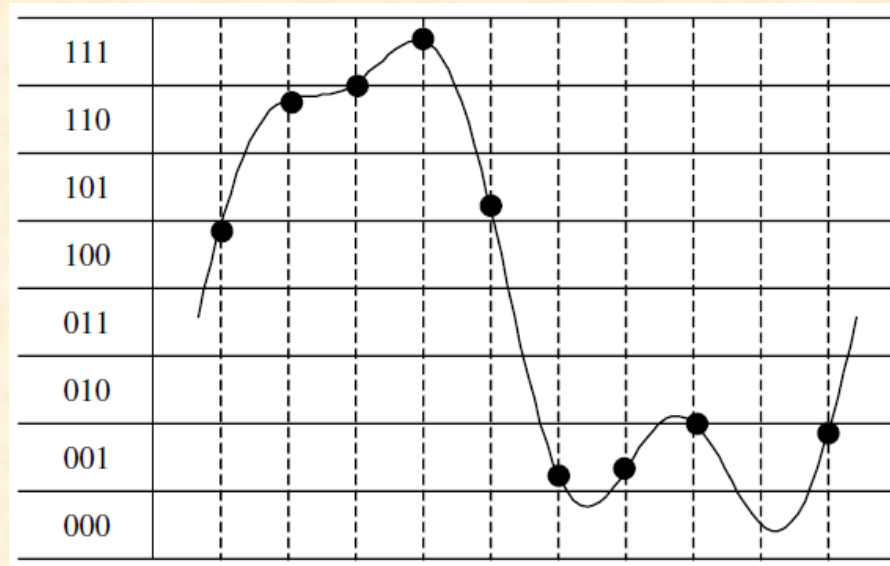
Example:

A 3-bit Analog-to-Digital Converter

- input voltage range: 0 to 10V
 - i.e. **infinite values** (in between 0V to 10V range)
- output:
 - possible output values: 0, 1, 2 7
 - i.e. only $2^3 = 8$ **values**

Quantizing and Encoding

- Approximates a continuous range of values and replaces it with a binary number
- Error is introduced between input voltage and output binary representation
- Error depends on the resolution of the ADC

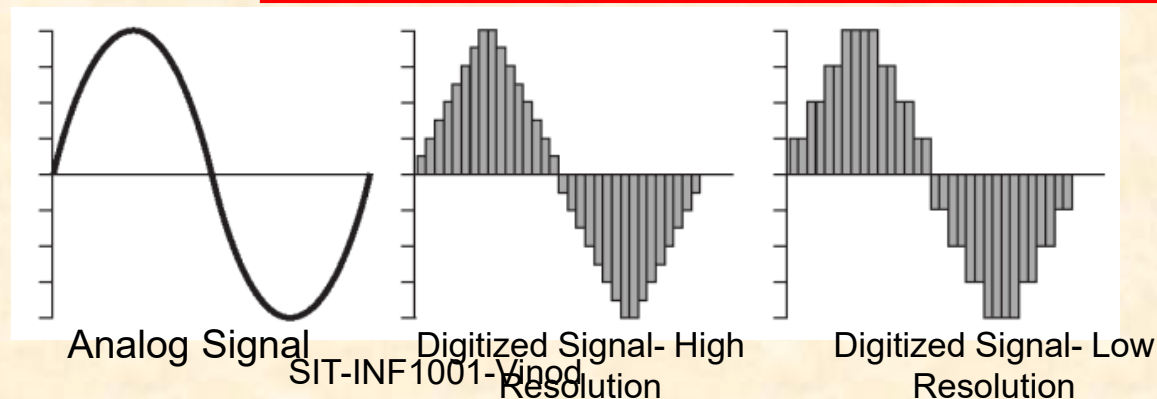


Resolution

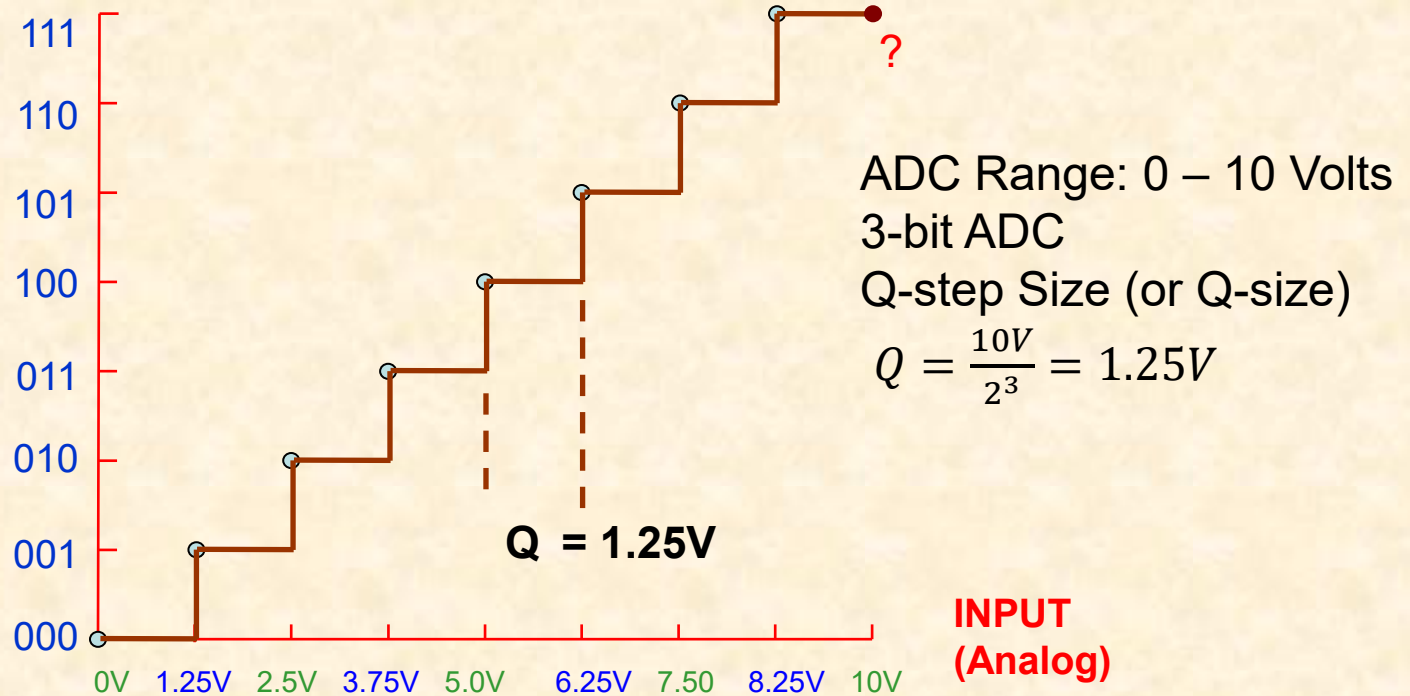
Increase in resolution improves the accuracy of the conversion

Resolution (bits)	Discrete states	Binary weight	Value of Q for 10 V FS
4	16	0.0625	0.625 V
6	64	0.0156	0.156 V
8	256	0.00391	39.1 mV
10	1024	0.000977	9.76 mV
12	4096	0.000244	2.44 mV
14	16384	0.0000610	610 μ V
16	65536	0.0000153	153 μ V

Minimum voltage step recognized by ADC



Quantization Size



Quantization Step-Size (Quantum) Q

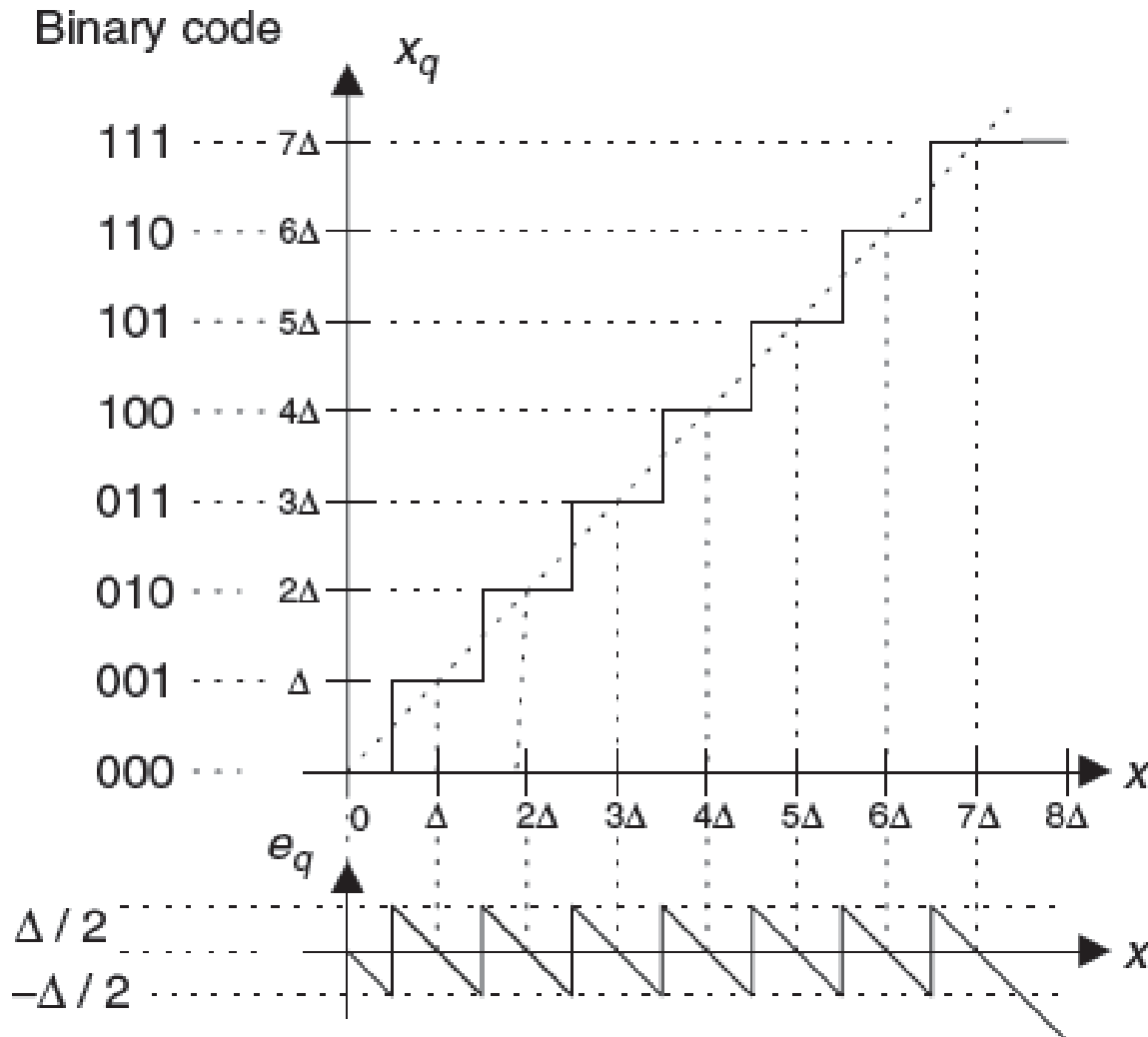
- the input voltage range for which its digital o/p value remains the same
 $\Rightarrow$ **uncertainty (error) in the actual input analog value**
- $Q = FS/2^n$ (where 'FS' is the full-scale value and 'n' is number of bits).

Quantization Error

The quantization error, e_q , is bounded by half of the step-size, that is,

$$-\frac{\Delta}{2} \leq e_q \leq \frac{\Delta}{2},$$

where Δ is the quantization step-size, or the ADC resolution in volts (also referred as $V_{\min}$, minimum detectable voltage) or the LSB value of the ADC.



Resolution of ADC

Can be expressed/described in terms of

- **number of bits** (n) of ADC
e.g. 8-bit, 10-bit
- actual **Quantization step-size** $Q = FS/2^n$
where FS = Full Scale input
e.g. 1.25V (for a 0V-10V, 3-bit ADC)
2.44mV (for a $\pm 5V$, 12-bit ADC)

Since $FS/2^n =$ **Least Significant Bit** size

- Q is also described as **1 LSB**
e.g. Error = $\pm \frac{1}{2} Q = \pm \frac{1}{2} \text{LSB}$

Note: 1 LSB error for 8-bit ADC = $1/2^8 \equiv 0.4\%$!

Note: 1 LSB error for 16-bit ADC = $1/2^{16} \equiv 0.002\%$

Example Problem

Q: 1 hour of Music is stored in a 2-channel CD sampled at 44.1 kHz using 16 bits per sample. How much storage space is required to store the music.

Sampling frequency = 44.1 kHz (means 44,100 samples in one second).

Total number of samples in 1 hour = $60 \times 60 \times 44,100$ samples.

Number of bits used to store one sample = 16 bits

Total number of bits required to store 1 hour music = $60 \times 60 \times 44,100 \times 16$ bits

Number of channels in CD = 2

Total bits for 2 channels = $2 \times 60 \times 60 \times 44,100 \times 16 = 5,080,320,000$ bits
(= 635.04 MB)

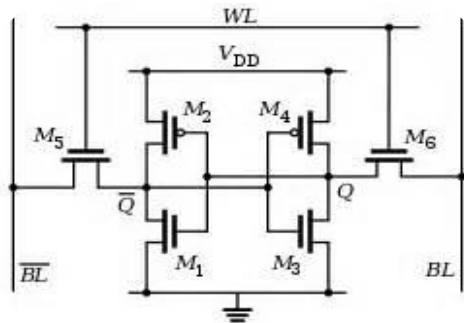
Q: How do you store several hours of music in a CD/digital storage device?

Compression (mp3)

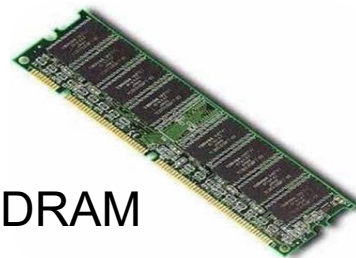
Storage of Digital Data – The Memory

SIT Internal

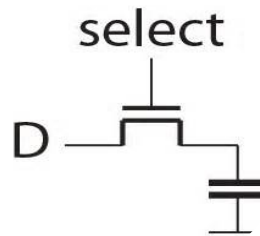
- Digital data can be stored in:
 - Transistor-based digital circuits storage systems such as SRAM, DRAM, ROM, FLASH
 - Magnetic and optical storage systems such as hard disk and CD/DVD



6 Transistors SRAM



DRAM



1 Transistor DRAM



SSD



Hard disk



CD/DVD

Transistor

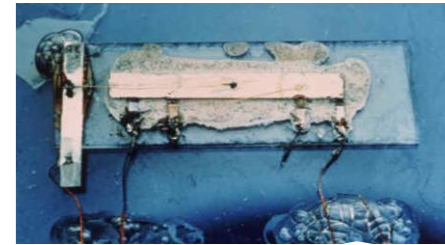
- A transistor is a semiconductor device used to amplify and **switch** electronic signals.
- Fundamental building block of modern electronic devices - **Integrated Circuits (IC)**.
- Inventors John Bardeen, Walter Brattain, and William Shockley were jointly awarded the 1956 Nobel Prize in Physics.



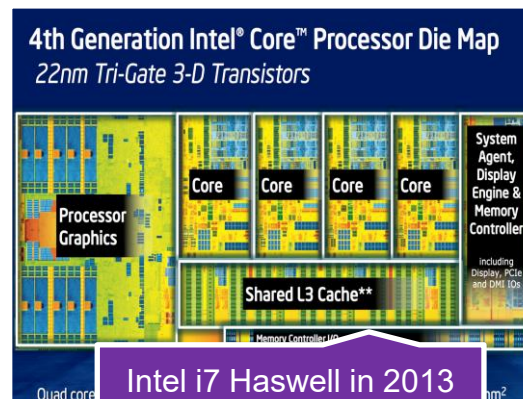
Transistor Invented in
Bell Labs in 1947



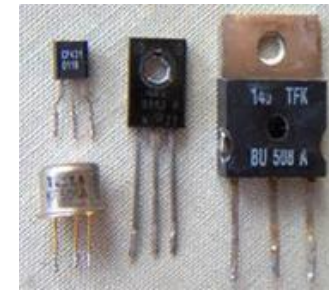
First Integrated Circuit invented in 1958
by Fairchild



Intel Pentium Pro in 1995
5.5 million transistors



Intel i7 Haswell in 2013
1.4 billion transistors



Types of Transistors

The Story of Silicon Valley (San Francisco)

SIT Internal

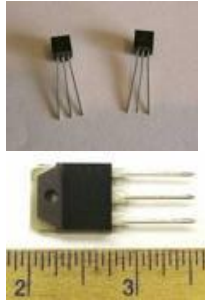
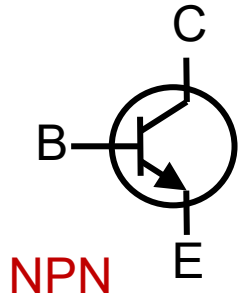
- ❖ Shockley moved to Stanford and founded Shockley Semiconductors in Palo Alto, the beginning of Silicon Valley.



- ❖ Shockley hired the brightest scientists and engineers (e.g. Gordon Moore and Bob Noyce, founders of Intel) who founded many companies after they left Shockley Semiconductors.

Transistor as a Switch (NPN Transistor)

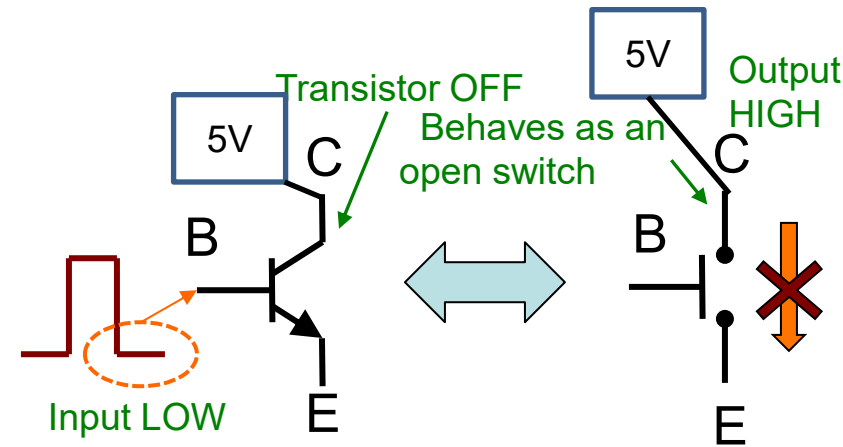
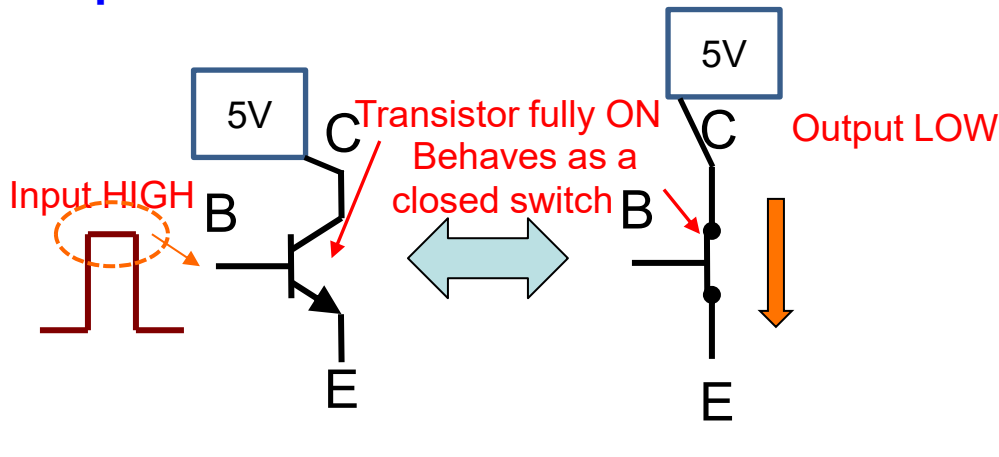
SIT Internal



The leads are labeled Collector (C), Base (B) and Emitter (E).

- The N and P refer to the material type that is used to construct the transistor. 'N' material has excess negative charges and 'P' material has excess positive charges.

Operation of Transistor as a Switch

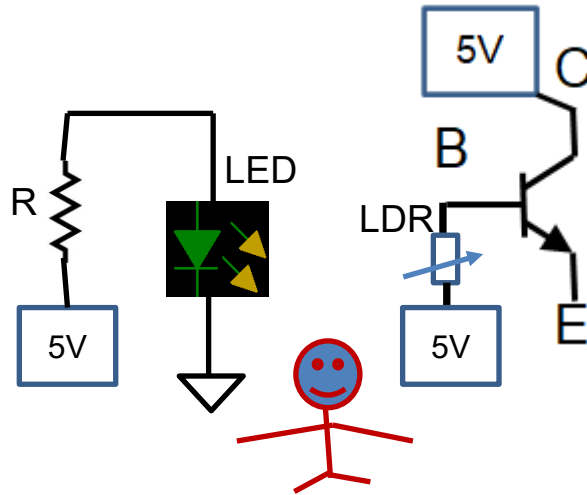


- Transistor is like a switch that can turn other switches on and off.
- We can use it to build up networks of switches all the way into combinational logic devices, digital memory, and full computers.

Transistor Switching – a Daily Life Application (7-Eleven store!)

SCENARIO #1 - NO OBSTACLE (PERSON) BETWEEN LED AND LDR

LDR (Light Dependent Resistor) – Resistance is very high when no light from LED (i.e., LDR is OFF) and Resistance is zero when light falls from LED (i.e., LDR is ON).

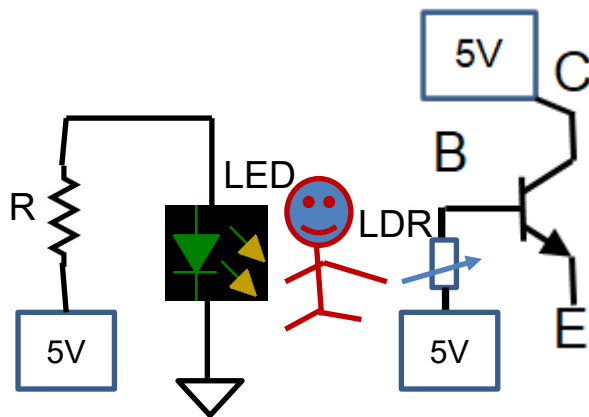


Light from LED falls onto LDR
 → LDR turns ON
 → Transistor Input becomes HIGH, Collector current flows fully
 → Output V_{out} goes LOW.

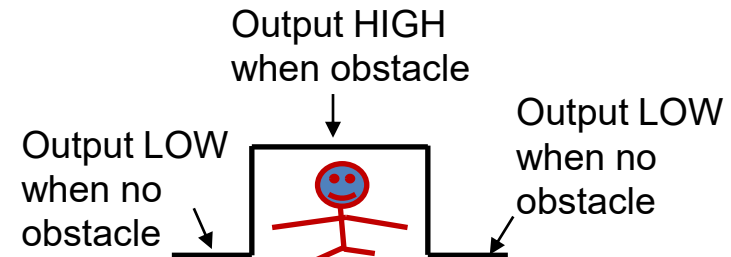


SBS Transit: Can you design a system counting vacant seats in upper deck?

SCENARIO #2 - OBSTACLE (PERSON) BETWEEN LED AND LDR



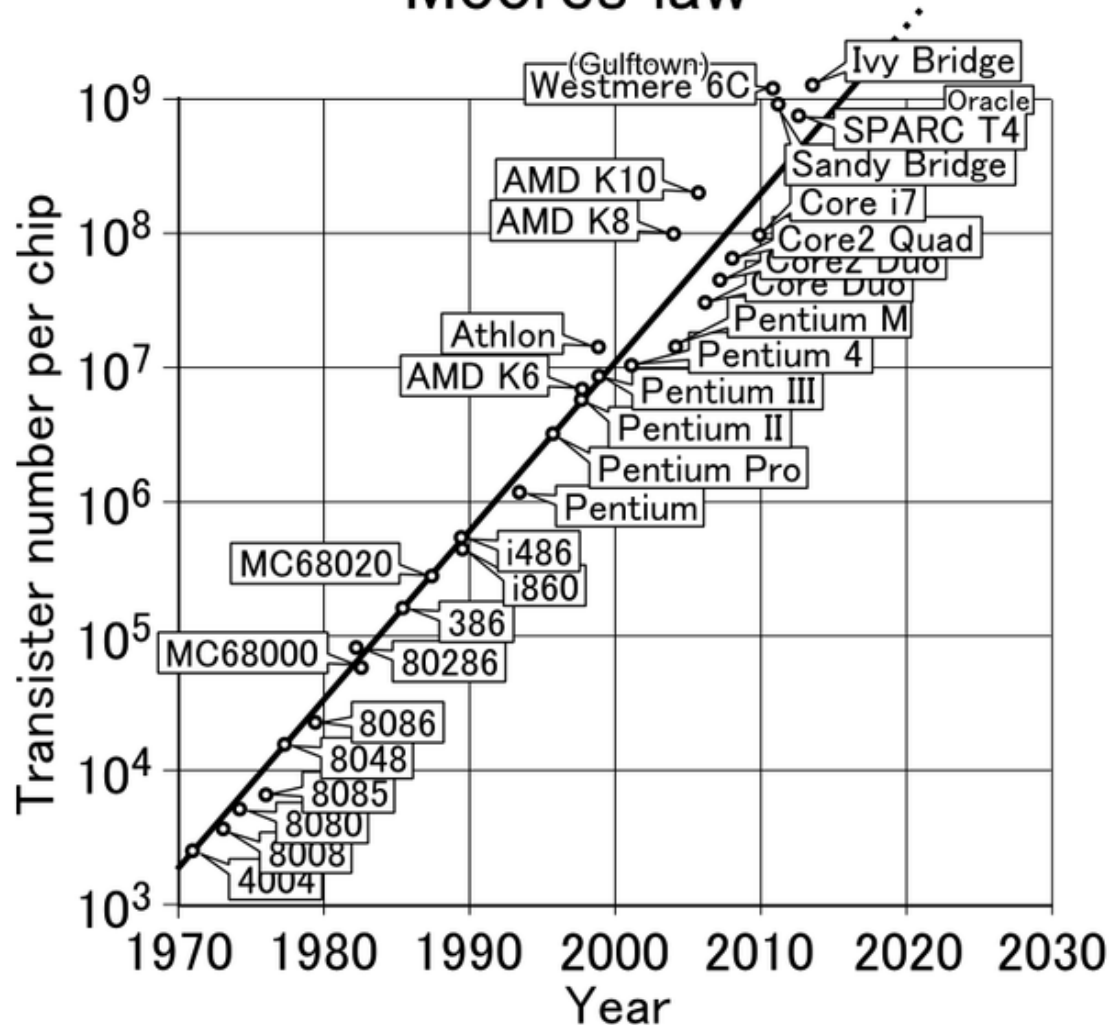
Light from LED does not fall onto LDR
 → LDR turns OFF → Transistor Input becomes LOW, No Collector current flows
 → Output V_{out} goes HIGH and triggers 'ting-tong'!



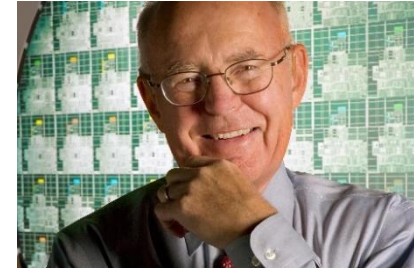
Think: How do you incorporate a counter that counts the number of persons walked in?

Moore's Law

Moore's law



<http://www.cringely.com/2013/10/15/breaking-moores-law/>



https://en.wikipedia.org/wiki/Gordon_Moore

Moore's Law

- Moore's law is the observation that, over the history of computing hardware, the number of transistors in a dense integrated circuit doubles approximately every two years.

Challenges

- Intel 4th Generation Haswell chip uses a manufacturing process to place 1.4 billion transistors 22 nanometers apart.
- Size of a transistor limit by the size of a single atom (e.g. A single phosphorus atom, by comparison, is just 0.1 nanometers across)

Summary

- Introduction to the digital world and ICT
- Evolution of computing machines
- Analog versus digital signals
- Binary Number Representation
- Signal Digitization (Sampling & quantization)
- Transistors and Integrated circuits, Moore's law